# Software Requirements Specification (SRS)

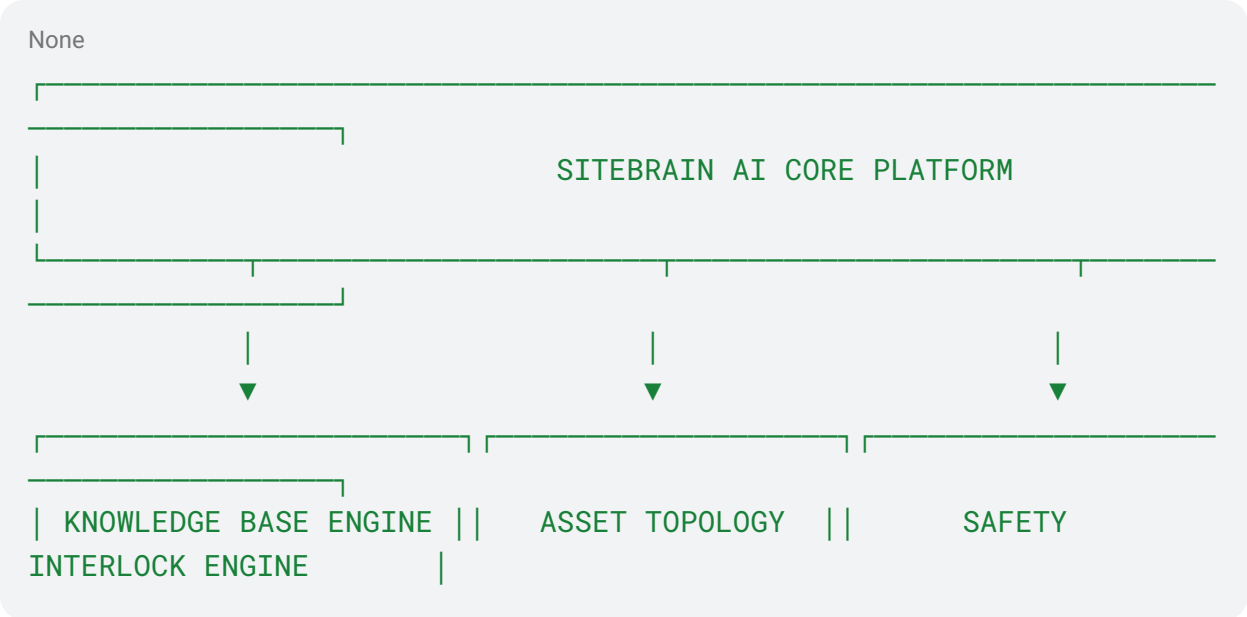
## Document Control

- **System Name:** SiteBrain AI — Unified Conversational Operations & Technical Escalation Copilot
  PRD - SiteBrain AI — BUU Operations Copilot V3.0
- **Target Sector:** Biogas Upgradation Units (BUU) Fleet Operations & Biomanufacturing
  PRD - SiteBrain AI — BUU Operations Copilot V3.0
- **Primary Owner:** Manojkumar Gothandapani · OPTIMAXX
  PRD - SiteBrain AI — BUU Operations Copilot V3.0
- **Document Version:** V4.5 (Full SRS & Functional Architecture Blueprint)

## 1. System Scope & Architecture Overview

SiteBrain AI is an autonomous, vendor-agnostic Industrial Operations Copilot designed to unify multi-modal knowledge retrieval, P&ID asset topology mapping, deterministic process safety interlocks, and context-preserving human escalation into a single operational interface.

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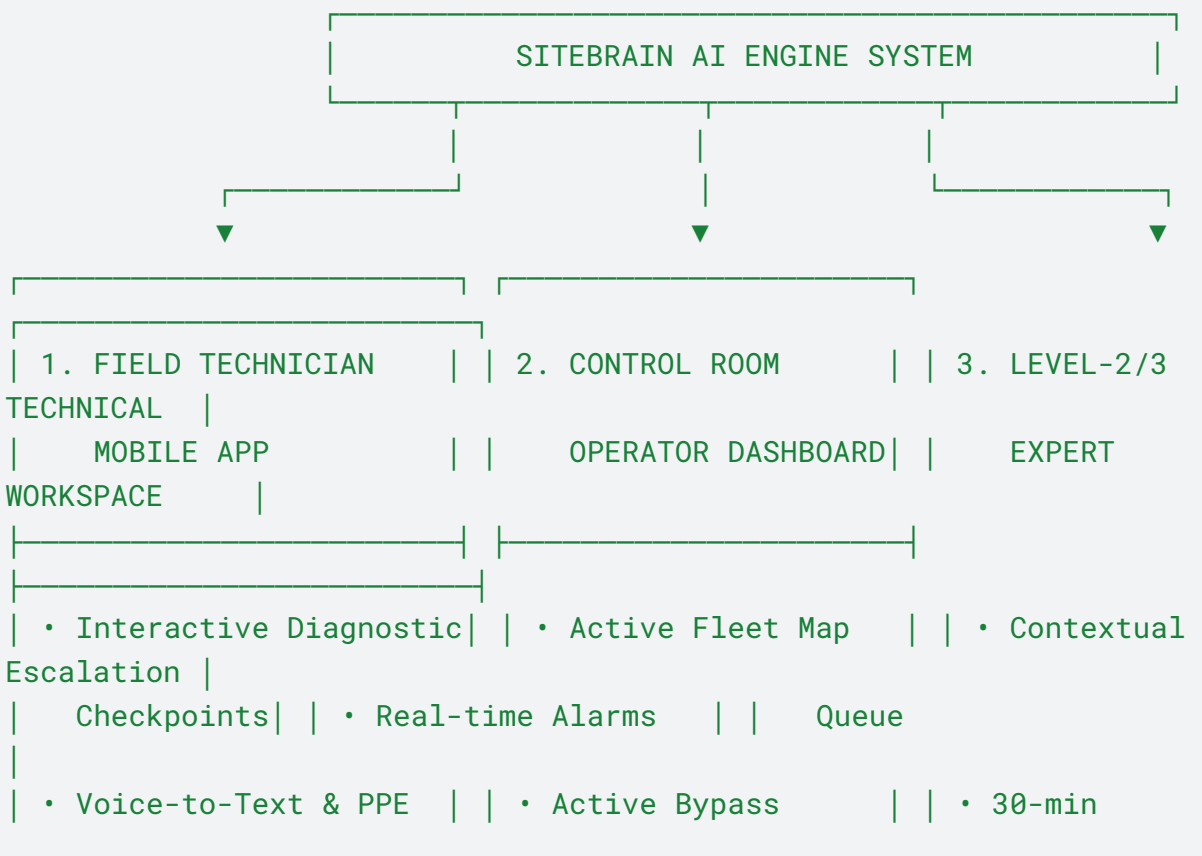


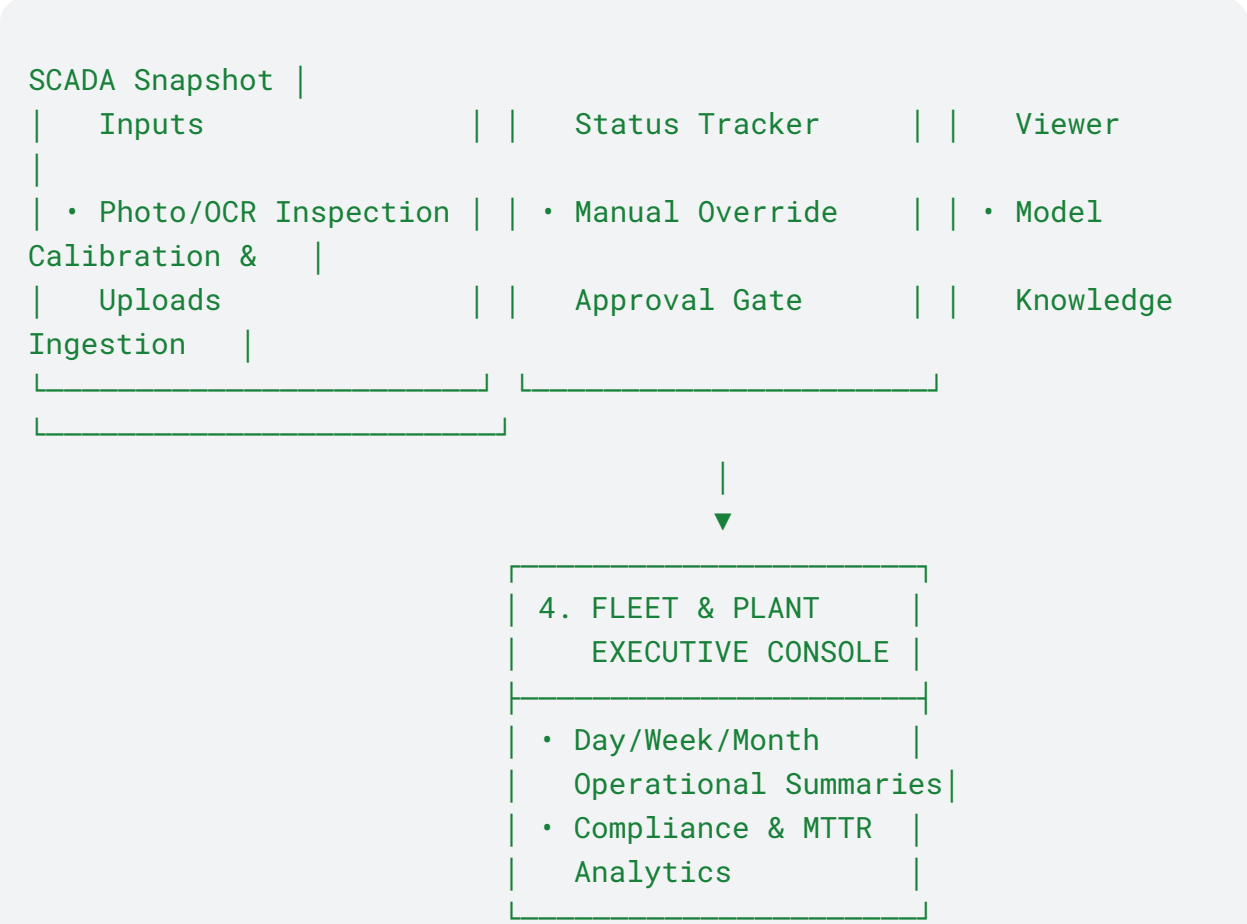
```
|  • OEM Manuals (OOM)  ||   • P&ID Stream   || • SIL-2 Control
Protection         |
|  • Maintenance SOPs   ||     Mapping       || • LOTO Override
Enforcement |
|  • Historical Logs    ||   • Sensor Nodes  || • Real-time Trip
Logic             |
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```

## 2. Multi-User Interface Specifications & Role-Based Access Control (RBAC)

The system enforces strict multi-tenant, role-based functional views across four primary operational profiles:

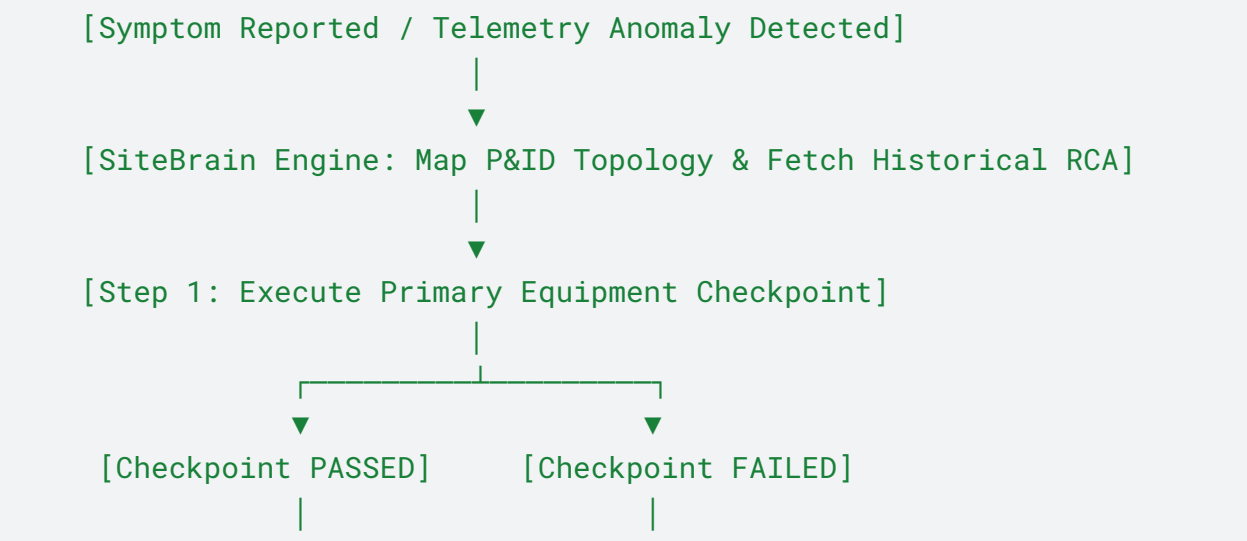
None

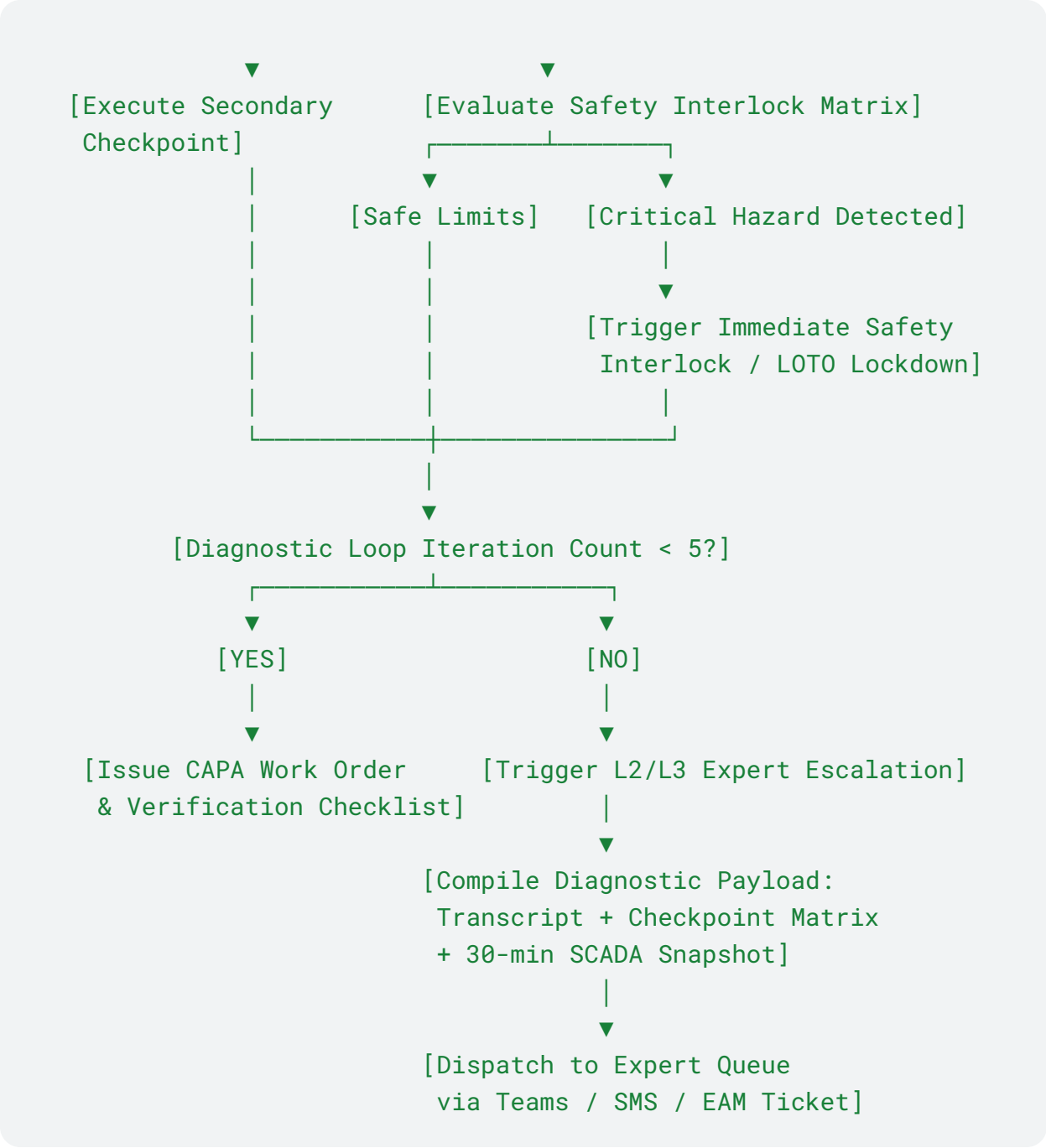





## 3. End-to-End System Process Flow & State Machine

None






# 4. Sub-System Diagnostics & Technical Escalation Protocols

## 4.1 H2S Scrubber & Pre-Treatment Diagnostics

1. **Symptom Ingestion:** Technician inputs differential pressure (dP) spike or downstream

sulfur breakthrough.

2. **Topology Scan:** Engine maps H2S Scrubber Vessel → Downstream Gas Filtration Skid → LP Gas Compressor Suction Vessel.
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3. **Primary Checkpoint:** Prompt operator to inspect top distributor mesh alignment and perimeter seal gap.
4. **Interlock Trigger:** If top distributor mesh failure or media channeling is flagged, issue an immediate interlock command locking downstream filter bypass valves in the closed position.
   PRD - SiteBrain AI — BUU Operations Copilot V3.0
5. **Resolution Generation:** Populate de-choking, media re-seating, and flushing protocols.

## 4.2 Technical Expert Escalation Handler

The system halts automated resolution loops and dispatches a high-priority payload when:

- Conversation turn count exceeds 5 diagnostic turns without root-cause confidence >85%.
- Physical condition inputs directly contradict real-time SCADA telemetry.
- Critical safety risks (e.g., solid particle ingress into gas compressors) are detected.
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**Standard Escalation JSON Payload Format**

JSON

```
None
{
  "escalation_metadata": {
    "ticket_id": "ESC-BUU-2026-0915",
    "timestamp_utc": "2026-09-15T18:00:00Z",
    "site_code": "GPS-BUU-04",
    "asset_id": "SCRUBBER-H2S-01",
    "urgency_level": "CRITICAL"
  },
  "diagnostic_summary": {
    "turns_executed": 5,
    "ai_confidence_score": 0.42,
    "primary_symptom": "Unstable pressure drop and particulate
migration suspicion",
```

```
    "failed_checkpoints": [
      {
        "checkpoint_id": "CHK-SCR-01",
        "description": "Top Distributor Mesh Alignment",
        "status": "FAILED",
        "operator_input": "Perimeter seal gap observed"
      }
    ]
  },
  "telemetry_window_30min": {
    "inlet_pressure_bar": [1.2, 1.22, 1.21, 1.45],
    "differential_pressure_dp": [0.05, 0.04, 0.01, 0.02],
    "filter_bypass_valve_state": "OPEN"
  },
  "escalation_target": {
    "channel": "MS_TEAMS_EXPERT_QUEUE",
    "assigned_role": "LEVEL_3_PROCESS_ENGINEER"
  }
}
```


## 5. Equipment Inspection Checkpoint & Safety Matrix

| Equipment Category | Physical Checkpoint | Normal Baseline Range | Fault Indicator | Mandatory Safety Interlock |
|---|---|---|---|---|
| **H2S Scrubber Vessel**<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Internal Mesh & Top Distributor Alignment<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | 100% perimeter contact; fasteners torqued<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Bed channeling; sudden dP drop; media carryover<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Lock downstream filter bypass valve in closed position.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |

| | | | | |
|---|---|---|---|---|
| **Gas Filtration Skid**<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Filter Element & Bypass Valve Position<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Bypass valve closed; dP<baseline limit<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Bypass open; high element dP<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Invalidate bypass execution unless secondary guard filter is active.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
| **LP Gas Compressor**<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Suction Vessel Strainer & Valve Assemblies<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Clean gas passage; uniform valve temperature<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Particle buildup in vessel; suction valve temp spike<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Trip compressor load line if solid particle risk is flagged upstream.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
| **Chiller & Dehydration Unit**<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Dew Point & Condensate Drain Function<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Gas temp 2◦C−5◦C; auto-drain cycling<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Gas temp >8◦C; water carryover downstream<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Halt feed to separation membranes if dew point exceeds limits.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |

| | | | | |
|---|---|---|---|---|
| **Membrane / PSA Upgrader**<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Permeate dP & Tail Gas Recycle Valve<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Purity >95% CH4; normal pressure ratio<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Methane purity drop; abnormal tail gas flow<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Divert off-spec gas to flare/recycle loop; prevent pipeline injection.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |

## 6. Project-Wise & Fleet Data Handling Engine

To support fleet operations across multiple Biogas Upgradation Units, data processing is partitioned into three isolation layers:

None

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│                          MULTI-PROJECT DATA ARCHITECTURE                                     │
│                                                                                              │
├──────────────────────────────┬──────────────────────────────┬────────────────────────────────┤
│ 1. TENANT & SITE             │ 2. DATA PIPELINE             │ 3. COLD-STORE &                │
│    ISOLATION LAYER           │    INGESTION LAYER           │    COMPLIANCE AUDIT            │
├──────────────────────────────┼──────────────────────────────┼────────────────────────────────┤
│ • Logical isolation by       │ • Real-time MQTT/OPC-UA      │ • Immutable append-only        │
│   `Site_ID` and              │   telemetry streams.         │   audit logs for all           │
│   `Project_Code`.            │ • Batch ingestion of         │   operator choices.            │
│ • Secure RBAC mapping        │   PDF OEM manuals,           │ • Model
```

```
retraining datasets  |
|   for field crews.      |   SOPs, and shift logs. |   anonymized
across sites.   |
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```

# 7. Operational Analytics Dashboards & Executive Summaries

## 7.1 Daily Operational Summary Dashboard

- **Primary Focus:** Immediate plant stability, open shift anomalies, and daily maintenance adherence.
- **Key Metrics & Layout:**
    - **Daily MTTR & MTTD Tracker:** Average time elapsed from symptom ingestion to root-cause confirmation.
    - **Active Anomaly Matrix:** Open diagnostic sessions categorized by severity (Low, Medium, High, Critical).
    - **Daily Shift Checkpoint Audit:** Total checkpoints executed by field technicians vs. passed/failed ratios.
    - **Interlock Event Log:** List of mandatory safety overrides or bypass blockages executed by the engine during the last 24 hours.

## 7.2 Weekly Operational Summary Dashboard

- **Primary Focus:** Component wear trends, recurring failure modes, and technical escalation efficiency.
- **Key Metrics & Layout:**
    - **Weekly Sub-System Failure Distribution:** Pareto breakdown of faults across H2S scrubbers, compressors, chillers, and membranes.
    - **L2/L3 Escalation Rate:** Percentage of total incidents escalated to human experts vs. those resolved directly by the copilot.
    - **Mean Time to Escalate (MTTE):** Average elapsed time required to compile and dispatch escalation context packages.
    - **SOP Compliance Index:** Percentage of maintenance activities completed using

the copilot's structured verification workflows.

### 7.3 Monthly Executive Summary Dashboard

- **Primary Focus:** Strategic fleet availability, asset protection ROI, and system reliability metrics.
- **Key Metrics & Layout:**
    - **Fleet Uptime & Production Reliability:** Total operational hours maintained across all deployment sites.
    - **Critical Incident Avoidance Index:** Quantified instances where interlocks prevented major equipment damage (e.g., zero particulate ingress events into compressors).
    - **First-Contact Resolution (FCR) Trend:** Month-over-month trajectory of self-service diagnostic resolutions (Target >80%).
    - **Knowledge Base Expansion Rate:** Total new incident resolutions and RCA packages indexed into the RAG knowledge store.

# 8. System Requirements: Hackathon MVP vs. Enterprise Production

| Architecture Layer | Hackathon MVP Requirements<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Enterprise Production Requirements<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
|---|---|---|
| **Data Ingestion** | Batch processing of PDF operating manuals, CSV incident logs, and SOP documents.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Continuous real-time telemetry streaming via MQTT / OPC-UA from SCADA and Historians.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
| **Context & Graph Engine** | Rule-based asset dependency lookup table (H2S Vessel→Filter→Compressor).<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Dynamic Graph Database (e.g., Neo4j) mapping physical P&ID relationships, valve nodes, and sensor tags.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |

| | | |
|---|---|---|
| **Diagnostic Logic** | Guided multi-turn QA forms driven by decision tree scripts.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Adaptive probabilistic fault tree inference engine combined with multi-modal LLM reasoning.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
| **Safety Interlocks** | Visual warning banners and mandatory checkbox sign-offs in chat interface.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | SIL-2 compliant automated interlock triggers executed via control system (PLC/DCS) logic.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
| **Escalation Output** | Auto-generated email/SMS digest containing conversation summary and diagnostic table.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Automated dispatch to PagerDuty/MS Teams and REST API integration with enterprise asset management (EAM/SAP).<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |
| **Interface** | Responsive web chat interface with inline checkpoint tables.<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 | Multi-platform deployment (Mobile app for field technicians, web console for control room, voice-to-text integration).<br><br>PRD - SiteBrain AI — BUU Operations Copilot V3.0 |

# 9. Functional Success & Reliability Metrics

- **Diagnostic Accuracy:** >90% accuracy in identifying correct root causes on initial diagnostic run.
  PRD - SiteBrain AI — BUU Operations Copilot V3.0
- **Safety Adherence:** 100% blocking of unsafe bypass or startup attempts via mandatory

interlock checks.



- **First-Contact Resolution:** >80% of routine field operational queries resolved directly by copilot without human escalation[cite: 10].
- **Escalation Speed:** Context payload compiled and dispatched to human technical experts in <30 seconds upon escalation trigger[cite: 10].
- **Context Preservation:** 0% loss of diagnostic state during human handover, eliminating repeat questions by technical experts[cite: 10].